

The Trading Behaviour of UK Institutional Investors

Aslihan Ersoy-Bozcuk^{a,1}, M. Ameziane Lasfer^{a,*}

^a City University Business School, Barbican Centre, London EC2Y 8HB, UK

Abstract

In this paper we analyze the trading patterns of a large number of institutional investors in the UK. We expect these investors to time their trades and the buy (sell) trades to lead to positive (negative) post-event returns. On average, share prices increase significantly on the announcement of both large buy and sell orders and before the sell orders, suggesting that institutional investors take advantage of the run-up in share prices before they sell their holdings. However, we document significant differences across different institutional investors in terms of the timing of the trades with insurance companies, fund managers and overseas investors following contrarian strategies while pension funds and banks follow momentum strategies. However, such strategies do not necessarily lead to positive returns in the post-event period.

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Key words: Institutional trading, Corporate monitoring; UK institutions

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* Corresponding author. Tel. (+44) (0) 171 477 8634, Fax (+44) (0) 171 477 8648, email: m.a.lasfer@city.ac.uk.

¹ Tel. +44 0411 80 2596, email: a.ersoy@city.ac.uk

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1. Introduction

The influence of institutional investors on the trading volumes of equity markets is widely recognised and well documented in the previous literature. For example, in 1989, it was estimated that nearly 70% of the trading volume on the New York Stock Exchange (NYSE) was due to trading by institutional investors and their member firms (Schwartz and Shapiro (1990)). There are also several theories on whether their trading moves markets away from the equilibrium level of prices. For example, Chan & Lakonishok (1993) show that institutional trades have a substantial effect on the prevailing prices in the market. Other studies show that in seller-initiated block trades mean temporary and permanent price effects are both present, whereas in buyer-initiated block trades only permanent price effects are observed (Kraus and Stoll (1972); Ball and Finn (1983); Ryngaert (1983)).

We use a unique data set that reports all block trades from 1993 to 1998 to test the hypothesis that institutional investors time their trading and the market expects good or bad news to follow from such trades. First, we analyse the pre-trade abnormal returns determine whether institutional trades are driven by herding behaviour. We then analyse the abnormal returns on the day the trades occur to assess the extent to which such trades affect the equilibrium share prices. Finally, we analyse the post-event abnormal returns to test the hypothesis that the share price impact is permanent. In addition, we take a different perspective from the previous literature and argue that, when institutional investors acquire large stakes in companies, they are expected to lead such companies to a better performance and, therefore, share prices of these companies increase in expectations of the outcome of such monitoring. In contrast a selling activity by institutional investors should result in a decrease in share prices because it will signal bad news and an end of blockholder monitoring.

We identified 8,590 buy trades and 8,136 sell trades undertaken by occupational pension funds, fund managers, insurance companies, banks, industrial and commercial companies and investment trusts. The average proportion of shares traded is 2.27 per cent ranging between 0.01 per cent to 57 per cent. For the buy

orders, we find that share prices increase by 0.27% over the [-1 to 0] period. In the pre and post event period none of the abnormal returns are significant at any confidence level. For the sell order, we find that share prices increase abnormally by 0.29% over the [-1 to 0] period. However, this abnormal performance started well before the event period. We show that prices increased by an average of 2.8% over the [-40 to -2] period.

We find however that institutional investors are not homogeneous in their trading activity. For example, fund managers and investment trusts appear to time perfectly their trades. In the 35-day pre-trade period share prices decrease abnormally by 0.65% before they buy and they increase by 3.1% before they sell. In addition, investment trusts do not appear to buy winners and sell losers as share prices carry on increasing by an average of 2.1% after the buy order while the post-abnormal performance of their sells is not statistically significant. In contrast, pension funds appear to lose on their trades. For example, in the pre-buy period [-40 to -6], share prices increase by 2.1% and, after they buy, prices decrease by 1.36%. On the event date 0, share prices increase for both buy and sell orders independently of the category of institutional investor. Our results suggest that institutional investors affect the short-term equilibrium share prices but some institutions are better than others in timing their trading strategies.

The rest of the paper is organised as follows. In section 1 we provide the theoretical background. In Section 2 we describe the data and methodology. In Section 3 we present the empirical findings. Conclusions are in Section 4.

2. Theoretical background

In this section we review the relevant literature, present the institutional framework and set up the hypotheses.

2.1 Review of the literature

A Why do investors trade?

There are several different factors that may lead an investor to trade. The trade itself can be a combination of the different investment styles (active vs. passive, value vs. growth, etc.) and order placement strategies (market vs. limit orders) available to

the investors. A buy or a sell transaction might merely be carried out for portfolio rebalancing purposes.

However, if an investor has a piece of information that s/he believes might move the company's share price upwards when publicly known, then s/he might be motivated to buy the company's shares. Conversely, it could also be that the investor has received adverse information about the company and is selling the stock to reduce his exposure. If this information relates to corporate governance issues, the selling activity suggests that institutional investors chose the exit route offered by liquid markets rather than the monitoring strategy, thus hindering effective corporate governance (Bhide, 1993). However, large blockholders, when faced with poor corporate performance, chose between selling the stock or intervening and that this potential conflict can actually be alleviated if stock markets are sufficiently liquid to allow the blockholders to purchase additional shares in order to use their power to influence (or even replace) incumbent management (e.g., Maug 1998). Thus increased stock market liquidity does not necessarily lead to a decline in monitoring activities.

This argument brings us to the corporate control aspect of owning large blocks of shares. Barclay and Holderness (1991) suggest that, block trades, even when they do not lead to the acquisition of the whole firm, are potential corporate control activities if they provide the buyer with sufficient voting power to 'have a say' in the composition of the top management team. They add that a mere 10-15% block ownership can bring about substantial corporate control. However, surely, not all blockholding will be motivated by the desire to influence management. It might also be the case that the block buyer is in pursuit of synergies through vertical or horizontal strategic integration with the company.

By the same token, not all sell transactions will be motivated by some adverse information. Although 'paper profits' are nice to have, a professional money manager will eventually want to sell the stock in order to realize these gains. Soon after this sell transaction, the same investor might buy shares in the same company but this time for a different client. Scholes (1972) disagrees and suggests that since investment companies and mutual funds invest considerable time and effort into research and maintain close contacts with brokers and underwriters, their move away from a stock is more than likely to be based on adverse private information¹. On the other hand, he estimates that a vast majority of estates, trusts, individuals, banks and insurance

companies sell for reasons other than possessing adverse information. He explains that these investors are not likely to closely follow the day-to-day operations of the firm and might sell the company's stock for a broad range of reasons. These range from "meeting tax obligations, making philanthropic donations or other distributions to legatees" (for an estate) to "consumption needs" (for an individual investor) (p.200). However, it is important to remember that Scholes' work is based on secondary distributions, which run by different mechanism than a simple buy or sell transaction. Also his data covers the period 1947 to 1965 for the NYSE stocks, which might make his results not only time-specific but also context-specific.

Whatever the motives might be behind the buying or selling of stocks, there is evidence of a move in share prices caused by the trade. The next section reviews the theory and empirical studies regarding the market response to trades.

B Market Response to Block Trades

In a simplistic framework, when a blockholding changes hands, this transaction should have no effect whatsoever on the firm value. After all, there is a buyer for every seller and given that these investors are rational and informed decision-makers, there should be nothing fundamentally wrong with the company. Therefore, we would not expect equilibrium prices to move in response to this trade. However, Chan & Lakonishok (1993) suggest the general opinion of the public is that institutional trading does occur frequently and involves considerable chunks of stock with an equally substantial effect on the prevailing prices in the market. Empirical literature agrees on the finding that in seller-initiated block trades, mean temporary and permanent price effects are both present, whereas in buyer-initiated block trades only permanent price effects are observed (Kraus and Stoll, 1972, Ball and Finn, 1983, Ryngaert, 1983). However, the measurement of block size, and the definition of a large block and of price effects differ considerably in reaching the above conclusions.

There are several different views on what exactly constitutes a block. Holthausen *et al.* (1987) suggest that any trade in NYSE that involves at least 10,000 shares is classified as a block. The market participants, however, refer to a 10,000-share trade as a "vanilla block", implying that "little or no price effect should be observed for a transaction of that size for large firms" (p.244). In the London Stock

Exchange, assuming that the delayed publication rules for trades can be used as a proxy for the Exchange's views on what constitutes a block then, we can say that a trade size greater than 6 times Normal Market Size (NMS) should be considered a block trade.²

In the literature, alternative measures of block size comprise; transaction volume of at least 5,000 shares (Holthausen *et al.*, 1987), at least 5 percentage of common stock represented by the trade (Barclay and Holderness, 1991) and a beneficial interest of 5%³ or more of a firm's outstanding shares (McConnell and Servaes, 1990, Bethel *et al.* 1998). A recent development in the literature has been the identification of trades as packages. Chan and Lakonishok (1993, 1995) define a 'buy (sell) package' as the investment manager's successive purchases (sells) of the stock for which a five-day no-trade period marks the end of the package.

These various measurements can dramatically affect the data on which these empirical studies are based on. Holthausen *et al.* (1987) report that when sampling the largest blocks in terms of the dollar value, they end up with the block trades involving the larger firms. However, when they sample the largest blocks based on the percentage of equity or the block volume relative to 'normal' daily trading volume⁴, they are left with the smaller firms⁵.

The theories on the impact of institutional trading on share prices are based on 3 main views. According to the first view, institutional trading increases the long run price volatility in the market by destabilising stock prices and causing them to move away from their fundamental values. This view assumes that institutional trading tends to involve a large volume of shares changing hands, which is accompanied by a large shift in share prices in response to the trade. In addition, the off-balancing effect is intensified by 'parallel trading' that may be common among large institutional investors. Lakonishok *et al.* (1992) quote a pension fund manager to explain this problem: "Institutions are herding animals. We watch the same indicators and listen to the same prognostications. Like lemmings, we tend to move in the same direction at the same time. And that, naturally exacerbates price movements" (Wall Street Journal (October 17, 1989)).

Parallel trading, also termed as herding, simply refers to the correlation that might exist between institutional trades. It could be caused by institutions interpreting other institutions' trades as conveying information to the market (Banarjee (1992);

Bikhchandani, Hirshleifer and Welch (1992); Shiller and Pound (1989)), agency conflicts in the performance evaluation of investment managers⁶ (Scharfstein and Stein (1990)) or by similar type of institutional responses to the same ‘exogenous signals’ (Lakonishok *et al.* (1992)). Yet Lakonishok *et al.* (1992) point out that not all herding is bad. They explain that herding may actually have a stabilising effect on stock prices and make the market more efficient, assuming that institutions trade in response to the same fundamental information and help quickly bring the prices to the new equilibrium level or counter some irrational expectations of individual investors.

However, supporters of this view defend their argument by putting forward the fact that institutions’ trading strategies are not always based on fundamental information about the firm. They point out that it is a well-known fact that contrarian strategies⁷ are being used in the fund management industry as well as various other short-term strategies based on technical analysis and feedback trading. Positive-feedback trading, also known as trend chasing, refers to buying winners and selling losers with the expectation that the current movement in stock prices will continue. This strategy is well researched in the behavioural literature (Andreassen and Kraus (1988)) and may lead to a superficial movement in the stock prices away from their equilibrium level (De Long *et al.* (1990); Cutler, Poterba and Summers (1990)). An added incentive in using this strategy is avoiding embarrassment when all goes wrong, in other words, ‘window-dressing’ (Lakonishok *et al.* 1991). Lakonishok *et al.* (1992) warn that not all positive-feedback trading is destabilising and it might achieve just the opposite in a setting whereby stocks under-react to news.

This brings us to the second view of institutional investment behaviour which posits that institutional investors are rational decision-makers who not only actively seek more information about the companies that they invest in but also have the advantage of utilising the services and supervision of advisers and fund managers which all help them gain a clearer picture of the fundamentals. Given this setting, supporters of this view would expect institutional investors to follow negative-feedback strategies by “buying stocks that have fallen too far and selling stocks that have risen too far” (Lakonishok *et al.* (1992), p.27).

The third and final view is one of a compromise between the above two extremes. It rejects the arguments developed by the two previous views and suggests that the structure, the objectives and the investment strategies of institutional investors

is so diverse that it would be a gross over-generalisation to pigeon-hole them as, for example, trend-chasers or negative feedback traders. It recognises the heterogeneity of institutional investors and suggests that their investment strategies might actually counterbalance each other.

However, these views alone do not directly explain the movement in stock prices associated with institutional trading as observed by the empirical studies and it is important to review the potential sources of the market response.

There are 3 main sources of the impact of trading as identified by the literature:

The Price Pressure Hypothesis: Also termed as the short-run liquidity hypothesis, it suggests that when investors buy or sell large volumes of stock it would be impossible to achieve this without moving the markets, regardless of how large or efficient to particular Stock Exchange might be. Scholes (1972) explains that as the size of the trade increases, it might be necessary to offer a “sweetener” in order to convince investors to buy the additional shares now available in the market (p.180). He suggests that these additional shares will only be held at a lower price than the prevailing price in the market, given the excess demand curve for shares is downward sloping. Holthausen *et al.* (1987) further explain that the price concession given by the seller of a large block has two components: compensation for inventory costs and compensation for search costs. The inventory costs may include a risk premium, the size of which may depend on the size of the block and the return variance of the stock (Ho and Stoll (1981)). The search costs on the other hand, are likely to be much lower in liquid markets. Likewise, an investor who initiates the purchase of a large block may need to pay a premium which reflects the difficulty of finding sufficient number of sellers willing to immediately part with their holding. This hypothesis suggests that the price effect is temporary and the transaction prices subsequently return to the equilibrium level (Demsetz (1968); Kraus and Stoll (1972); Stoll (1978); Ho and Stoll (1981). Dann, Mayers and Raab (1977) report that this “rebound” can be observed, on average, within 15 minutes subsequent to the transaction.

The Imperfect Substitution Hypothesis The substitution hypothesis is based on the assumption that investors determine their future consumption streams by selecting different combinations of risky assets to include in their consumption-investment program. The risky asset is just one of the several alternatives available to the

investors and is priced in a way that the expected return on similar assets will resemble each other. Therefore, if a particular asset promise higher expected returns simply due to an increased availability in the market, the arbitrage opportunity will quickly be dissipated by market participants, and the price will move accordingly. As Scholes (1972) explains, “Since assets are substitutes in investor portfolios, the pure price effects [of investor purchases and sales] must be very small” (p.182). According to this hypothesis, we would expect the drop in the stock’s price to accommodate a large-quantity sell to be negligible. However, the imperfect substitution hypothesis suggests that if there are insufficient close substitutes for a particular firm’s stock, a seller might be faced with a downward-sloping demand curve, which will necessitate discount in stock price for the transaction to take place. Likewise, a buyer might be faced with an upward-sloping supply curve which will mean that for the large transaction to occur a premium will be necessary. This hypothesis predicts a permanent price effect⁸ or at least a slower price ‘rebound’ than that of the price pressure hypothesis.

Information Hypothesis The information hypothesis suggests that the purchase or sale of securities might convey information the market that may translate into a movement in stock prices. In this case, the markets are responding to the additional information signalled by the trade itself. It also suggests that, due to the hefty costs of looking for information valuable enough to help the investor beat the market, we would expect the seller of a large block of shares to have more information than a seller of a trivial quantity. That is, we would expect large block trades to include a greater amount of information than a trade carried out merely for portfolio-adjustment purposes. Hence, selling a large block of shares is likely to cause a downward pressure on the stock price that is not just a ‘sweetener’ but rather a permanent price adjustment. The permanent price effect will take place even when there are close substitutes to the firm’s stock which will lead to perfectly elastic demand curves (Kraus and Stoll (1972); Scholes (1972); Mikkelson and Partch (1985)).

The dilemma that remains is that, we would expect an informed block-seller to believe that the stock is over-valued, whereas we would expect the informed block-buyer in this transaction to believe that the stock is under-valued. However, they cannot both be right. Scholes (1972) explains that the information effect of a large block transaction will depend on the identity of the buyer or the seller since certain

categories of investors (such as insiders) are expected to have more information about the company than others. Many studies use the size of the transaction as a proxy for the information conveyed by the trade. However, Kyle (1985), in his rational expectations model, hypothesises that informed traders are more likely to maximise their trading profits by unfolding their trades gradually in the presence of liquidity (noise) traders. Chan and Lakonishok (1993) present evidence consistent with this view and report that although their sample of transactions comes from large money managers, the size of a typical trade is surprisingly small. They interpret this finding by suggesting that the large money managers “trade strategically to reduce the influence of short-run liquidity costs or information effects” (p.177).

C *Empirical Evidence*

Chan and Lakonishok (1993) analyse 1,215,387 transactions made by 37 large money management firms between 1986 and 1988, in the NYSE and the AMEX. They report that most of these transactions involve the largest stocks, which is a finding consistent with that of Lakonishok, Shleifer, Thaler and Vishny (1991). In calculating the effects of the trade (and classifying them as permanent and temporary) they follow Holthausen *et al.* (1987), however, they report a smaller price impact for both purchases and sales of stock. Buy transactions yield a principal-weighted average price increase of 0.22%, while sell transactions yield a price decline of 0.14%. They suggest the market response from the opening price to the trade price can be explained by short-run liquidity costs, prior release of information and positive-feedback trading by money managers. They find an asymmetric response between purchases and sales, consistent with Holthausen *et al.* (1987), Kraus and Stoll (1972), and Keim and Madhavan (1993). They report that market capitalisation and relative trade size play a role in the market impact of a trade; larger permanent price changes are associated with the purchases involving smaller firms. However, the most dominant influence on the price impact of trades turns out to be the identity of the money manager.

When Chan and Lakonishok (1995) modify their definition of block trades as multi-day packages, they report higher price impact associated with institutional trades. They find that the principal-weighted price change from the open to the close on the trade date is 1% for buy and -0.3% for sell-packages⁹. They also find that the asymmetry between the price impacts of buys and sells still holds.

On the other hand, Barclay and Holderness (1991) analyse 106 negotiated trades of at least 5% of the common stock from 1978 through 1982 and report average abnormal stock price increases of 16% associated with the initial public announcement of the block trade. They find even higher abnormal returns when the blockholder gains control of the firm, faces no resistance from management for attempting to influence corporate policies and eventually fully acquires the firm. They suggest that the information effects alone would not be able to produce these results and they attribute the positive abnormal returns to the specific skills and incentives of the block purchaser, as well as more valuable managerial and monitoring skills. However, their results may not be directly comparable with other studies due to the fact that they focus only on negotiated block trades. They acknowledge that, in their sample, a block trade does not change the concentration of ownership and the only thing that changes with the trade is the identity of the blockholder. In fact, none of their block trading parties was an institutional investor, possibly due to the legal requirements on diversification imposed on them.

Finally, Lakonishok *et al.* (1992) adopt a slightly different approach to identifying the institutional trades and look at the changes in the end-of-quarter portfolio holdings of 341 institutional money managers (managing the funds of 769 all-equity tax-exempt funds, most of which are pension funds) between 1985 and 1989. Their results suggest both the stabilising and the destabilising view on institutional investors are inaccurate. They provide evidence contrary to the popular belief that a large change in institutional excess demand is the driving force behind stock price movements. They find some evidence of herding and positive-feedback strategies in smaller stocks. However, they point out that most of the holdings of the institutions in their sample are concentrated in large stocks. They conclude that “institutional investors pursu[e] a broad diversity of trading styles that, to a large extent, offset each other” (p.24).

D Purchasing Blocks of Shares: A Route to Activism?

It is a known fact that, in the 1990s, activist investors purchase significant blocks of shares and exert influence over company policies. This movement has not been limited to only large institutional shareholders such as CALPERS and other public pension funds, but we have also seen individuals such as Carl Icahn, Kirk

Kerkorian, Bennett LeBow and Bob Monks use their block shareholding to bring about operational, financial and governance changes in large corporations. In fact, with the rise of block share purchases, the frequency of hostile takeovers and leveraged buyouts as a means of disciplining managers has declined substantially (Maug (1998); Bethel *et al.* (1998)). We now see a trend towards trying to gain higher portfolio returns through shareholder monitoring¹⁰ and governance related activities.

Poorly performing firms are potential targets of activism. Investors who own large blocks of shares may attempt to discipline managers by mounting a proxy contest, initiating adverse publicity, and in extreme cases, replacing management or taking over the firm (Butz (1994); Morck, Shleifer and Vishny (1989); Shleifer and Vishny (1986); Manne (1965)). By this way, the investors can be able to implement changes in corporate policy in order to improve firm performance and create shareholder wealth.

We would expect investors to purchase blocks of shares when the expected benefits of doing so outweigh the expected costs (Demsetz & Lehn (1985)). Bethel *et al.* (1998) explain that the potential benefits of holding a block of shares in a firm are mainly in the form of capital gains and the dividends received given that the blockholder can force policies upon the managers in order to improve company performance. The potential costs, on the other hand, are due to the loss of portfolio diversification and the allocation of resources to monitor management, mount proxy contests and ward-off potential legal challenges that may arise as a result of the blockholding. Bethel *et al.* (1998) add that managers can adopt defensive mechanisms that serve to diminish the net benefit of block share purchases and forcing changes in corporate policies.

They identify two main categories of defensive mechanisms. The first category works by “diminishing blockholders’ voting power, thereby preventing large investors from using proxy contests or otherwise exercising their voting power to pressure managers to change corporate policy” (p.608). Dual class share structures and employee share ownership schemes (ESOPs) are examples of this type of defensive mechanism. The second category works by creating “legal obstacles, thereby raising the cost of launching takeovers” (Bethel *et al.* (1998) p.608; Butz (1994); Shleifer and Vishny (1986)). Examples of this type of mechanism are anti-takeover charter amendments and re-incorporating in states with anti-takeover statutes.

Jarrell and Poulsen (1988) have reported negative share price effects on the announcement of dual-class share structures. However, the evidence for ESOPs is not as straight forward. Despite the fact that ESOPs are widely recognised as a means of facilitating performance-based incentives to employees, it is also argued that they may imbalance the voting structure and leave more power in the hands of incumbent management (Bethel *et al.* (1998)). In empirical studies, ESOPs have indeed been associated with decreases in firm value (Chang and Mayers (1992); Gordon and Pound (1990)) and decreases in operating profitability (Mikkelsen and Partch (1994)). As for anti-takeover charter amendments, negative stock price reaction to its announcement has been reported (Jarrell and Poulsen (1987); Ryngaert (1988); Malesta and Walkling (1988)). However, there is inconclusive evidence on whether anti-takeover amendments, such as supermajority provisions, classified board provisions, poison pills and preferred stock authorisations, can actually deter takeovers. Finally, negative average returns have been reported for firms incorporated in states with anti-takeover statutes in effect (Karpoff and Malesta (1989); Szewczyk and Tsetsekos (1992)).

The question remains whether we would observe block purchases as a route to shareholder activism more frequently, if it were not for the defensive mechanisms. However, we do know that purchasing blocks of shares can actually be a preferred route to activism by looking at the changes that occur in the firms subsequent to the block trade.

A number of studies in the literature have reported empirical evidence on the operational, financial and governance changes that follow block trades. Barclay and Holderness (1991) find that top manager and director turnover following the trades is far higher than the average level you would expect in a public corporation. 33% of the firms in their sample replace their CEOs within a year and only 26% have their chief executive and chairman unchanged within 2 years following the block trade.

Bethel *et al.* (1998) report results consistent with those of Barclay and Holderness (1991) and find that CEO turnover of firms which have experienced an activist¹¹ block purchase has increased from 10.7% in the two years before the block purchase to 22.3% in the two years after the block purchase. They also report that block share purchases by activists are followed by increases in divestitures and share repurchases and decreases in mergers and acquisitions. They inquire whether these

corporate changes lead to improvements in firm performance and find evidence in support of a statistically significant improvement in ROA¹² (and also industry-adjusted ROA) in the second and third years after the activist block purchases.

2.2. *The London Stock Exchange*

The London Stock Exchange has a number of features that differentiates it from the other International Exchanges. This section attempts to summarise some of these features that may be relevant within the context of this study.

A *Characteristics*

After the ‘Big Bang’ in October 1986, several important changes have occurred in the London Stock Exchange (LSE). One of these changes has been the introduction of a system whereby competing market makers sustain a continuous presence in the market and act as counter party to equity transactions. This represents a substantial move towards a more order-driven system.

The LSE also has a policy of ensuring the proper checks and controls are in place in order to protect investors, while at same time trying to avoid making these rules and regulations so rigid that it may scare-off companies who want to achieve a listing¹³. The ability and the relative ease of institutional investors in buying and selling large blocks of shares in a single transaction is actually one of the factors that bring a competitive advantage to the LSE when compared with other stock exchanges (Board and Sutcliffe, 1995). Board and Sutcliffe (1995) explain that this system requires the market makers to take quite substantial inventory positions, which increases the risk that they are subjected to. Therefore, market makers demand protection against this additional inventory risk. Protection can take several forms some of which can be delaying the publication of a large trade or the development of Inter Dealer Broker Systems (IDB) which provides a private, order-driven method by matching buys and sells and allowing the surplus to be traded among the other market makers.

B Delayed Publication

Despite the obvious advantage of offering protection to market makers, delayed publication can in fact be detrimental to the transparency of the market due to the information asymmetries that arise among the market participants.

Since the Big Bang the delayed publication policies for unusually large trades at the LSE have changed quite a bit. Prior to January 1991, although the volume of trades was published immediately, the publication of prices could be delayed for up to 24 hours for alpha stocks over £100,000 and 3 minutes for all other trades. From 14 January 1991, the publication of price and volume for trades larger than 3 times Normal Market Size (NMS)¹⁴ could be delayed for up to 90 minutes, while the delay allowed for all other trades remained the same. On 13 December 1993, an additional delay was allowed on top of the previously prevailing level and the market makers could chose to delay publication of trades larger than 75 times NMS from 90 minutes to 5 business days, or until 90% of the position has been unwound, whichever is sooner.

On 1 January 1996, in a move towards increasing the transparency of the market, the delayed publication of trades between 3 and 6 times NMS was abolished. Also, all IDB trades now had to be published with a delay of at most 3 minutes. Additionally, the delay period for trades greater than 75 times NMS was brought down from 90 minutes to 60 minutes (the minimum requirement).

Board and Sutcliffe (1997) report that subsequent to the delayed publication changes in 1996, there has been a decrease of 43% in the value of all trading subject to publication delay. They also find a 75% jump in the proportion of trading published immediately. They conclude that we can now expect to see the information effects of trades between 3-6 times NMS swiftly compounded in the stock prices¹⁵.

C Disclosure Requirements

On 18 September 1993, amendments, affecting both listed and unlisted companies, to the Companies Act (1985) (CA85) came into force in order to reflect the influence of the EU Transparency Directive 88/627/EEC. The regulations apply only to share capital with voting rights intact and require that any equity stake of at least 3% in a public company or an equity stake that no longer exceeds this threshold must be communicated to the company in the form of a written notification within two

days of the change. Any increase or decrease in the stake (above the 3% threshold) that causes the total to move up or down to the next % point necessitates a new notification. The company then needs to make appropriate changes to its share register within a period of 3 days subsequent to the date of receipt of the notification. Additionally, listed companies are required to inform the Company Announcements Office (CAO) of the London Stock Exchange as soon as they receive such notification. The information reported to the CAO needs to contain, among other details of the change in shareholding, the date of receipt of the notification and, if known, the date of the transaction.

For both the executive and non-executive members of the board of directors, however, any interest and any subsequent changes has to be disclosed and reported to the CAO in due course, with the date of the disclosure as well as the date of the transaction.

3. Data and Methodology

The data on changes in shareholding was collected from Extel Financial-Company Research. This database provides all the news items disclosed by the companies to the Regulatory News Service (RNS) Division of the Company Announcement Office at the London Stock Exchange. The disclosure requirement originates from the Continuing Obligations Section of the Listing Rules (Yellow Book). Disclosure requirements regarding the changes in the shareholding makes reference to the CA85 for the specific percentage change figures to be used as guidelines.

Company Research provides the news items exactly as they were submitted by the company to the RNS and subsequently disseminated by the RNS to all listed and member firms on-line real-time. The news item 'Shareholding in Company' is given as text (a short paragraph) with the date of the announcement (which is also the date that the company becomes aware of the change, by the mechanism explained in Section 5.3 above), name of the shareholder and the new percentage of share stake held. For nearly all 'Shareholding in Company' news items, the date of the transaction was not given.

There were several potential problems involved in this presentation of the news items. Firstly, it does not have a standard wording format, which necessitates individually sifting through the paragraph for the data needed for the study.

Secondly, it only gives the current percentage of shares held by the shareholder and does not specify the amount of the change from the previous level. This problem was overcome by collecting data for the year (1992) preceding the sample period (1993-1998) and computing the percentage change each announcement represents when compared with the announcement just before it.

Thirdly, the same shareholder's name appeared in several different formats making the comparisons unreliable. For example, the same fund manager can appear under the names: "PDFM", "Phillips and Drew Fund Management", "Phillips and Drew Fund Management Limited", "Phillips and Drew Fund Management Ltd", "Phillips and Drew". When you add to this list all the possible combinations with spelling errors (such as a "Philips", "Druw", "Manegement", etc.), it becomes virtually impossible to make the computer recognize that these actually are the same companies. Therefore, all the shareholder names were reviewed and replaced by code numbers. There are 1,504 individual institutions identified by the study. Finally, there was also variation in Company Names which was again solved by giving each company in the sample a unique number.

All news items titled "Shareholding in Company" was extracted from Company Research from 1993 through to end of 1998 and was rearranged as described above. When it is an individual holding the shares, the news items do not specify whether the individual is a director or not. Given that the sample is all listed companies in the London Stock Exchange, it was not feasible to try and check each individual's identity. Therefore, all news items with individual's names were dropped from the analysis.

After eliminating all events with unavailable or incomplete data for the full estimation cum event windows, the final sample consists of 16,726 events for 876 companies (including currently extinct companies) from 1993 through 1998.

Brown and Warner (1985) Event Study Methodology was used to compute Ordinary Least Squares (OLS) Market Model, with the parameters (α and β) calculated from the 250 day estimation window (event days: (-290,-41)). The event period consists of the 81-day period around the announcement day, day '0'. For

hypothesis testing, the t-statistic was computed by dividing the excess return by its estimated standard deviation. The Scholes and Williams (1977) procedure was used to correct for non-synchronous trading.

The FTSE-All Share Index was used to compute the return on the market (R_m). The company share prices were adjusted for capital changes and the dividend-adjusted, observed simple arithmetic returns (R_i) were calculated for each security.

4. Empirical Results

4.1. Descriptive Statistics

There are a total of 16,726 events, with 8,590 (51%) buy transactions and 8,136 (49%) sell transactions. The number of buy and sell trades in each year is presented in Panel A in Table 1. Despite the increasing trend in the number of trades towards the end of the sample period, there is no evidence of clustering.

[Please insert Table 1 about here]

Also in Table 1 (Panel B) we see that, the average size of the trade, as measured by the percentage of share capital traded, ranges between 1.99% (in 1998) and 3.10% (in 1994), while the median ranges between 0.93% (in 1998) and 1.42% (in 1993). The average £ value of the trade ranges between £9.5 million (in 1998) and £17.2 million (in 1995), while the median ranges between £1.1 million (in 1994-1995 and 1997-1998) and £1.3 million (in 1993). Therefore, the size of the trades is smaller than that reported by most of the previous literature on block trades.

[Please insert Table 2 about here]

Table 2 presents the descriptive statistics separately for all trades, for buys and for sells. The average % of share capital traded is greater for sell transactions (2.58%) than for buy transactions (1.97%), which also holds true when comparing the medians. Accordingly, the average % of shares held by the institutional shareholders immediately after the trade is smaller for sells (6.43%) than buys (10.11%), which is what we would expect. The average market capitalisation of companies that were the subject of a buy trade (£605 million) is similar to those of a sell trade (£628 million). The average £ value of the trade is greater for sells (£13.8 million) than for buys (£10.8 million).

[Please insert Table 3 about here]

Fund Managers are by far the largest category of shareholders according to the number of trades carried out during the sample period. As can be seen in Table 3, they account for 41% of all buy trades (3528 out of 8590) and 39% (3153 out of 8136) of all sell trades. Another striking picture in this table is the dominance of the Overseas Institutions who account for 14% of all buys and 10% of all sells.

4.2. *Abnormal Returns*

Table 4 provides a summary of the Abnormal Returns on each day of the event period for the buy and the sell transactions. Buy transactions experience statistically significant positive abnormal returns of 0.13% and 0.14% on event days -1 and 0 , respectively. The sell transactions on the other hand, experience consecutive positive abnormal returns starting from day -7 through to day 0 . The highest one-day abnormal return of 0.35% is achieved on day -2 . A graphical presentation of this table is provided in Figure1.

[Please insert Table 4 and Figure 1 about here]

Figure 2 depicts the Cumulative Abnormal Returns (CARs) for event window $(-40, +40)$. For buy trades, the CARs are negative until day 1 . The increasing trend starts from day -3 and continues until day 8 . For sell trades, the CARs follow a general upward trend from the start of the event window, with bigger increments between days -4 and 0 and they fluctuate around a 2.75% level after the announcement day.

[Please insert Figure 2 about here]

There seems to be some evidence of a “buy under-valued, sell over-valued” strategy being followed. For sells, there is no downward pressure on prices and there is even some positive post-announcement drift, which contradicts with what short-run liquidity hypothesis would suggest and supports the substitution hypothesis. This evidence is consistent with Chan and Lakonishok (1993) who explain that money managers might be involved in strategic trading in a way that will minimise the short-run liquidity and information effects.

For buys, however, it could be the buying behaviour that is driving prices upwards from day -3 and a positive post-announcement drift can be observed even up to day 8 . There seems to be a price reversal from day 8 onwards. Therefore, the positive abnormal returns just before announcement could be due to liquidity effects

and the positive abnormal returns just after the announcement could be due to information effects, both of which are temporary effects.

The results for buy trades are consistent with the findings of Chan and Lakonishok (1995). The CARs they have reported for buy trades (equal-weighted) follow a similar pattern of negative CARs just before the commencement of trade package¹⁶, cutting the x-axis (0%) half-way into the trade package and yielding positive CARs afterwards. The only difference is that they do not find evidence of a price reversal, whereas we find negative CARs after day 33.

Also, the results for sell trades are somewhat similar to that of Chan and Lakonishok (1995). The CARs they have reported for sell trades (equal-weighted) also seem to follow a run-up to day 0, although of a much smaller magnitude (0.37%). They interpret this finding as consistent with prior evidence (Lakonishok and Smidt (1987)) that “volume (and hence both buying and selling activity) tends to rise after increases in the stock price” (p.1158, Chan and Lakonishok (1995)). However, they find a drop in CARs at the start of the sell package which recovers on the last day of the package. This rapid rebound seems to be in support of the price pressure hypothesis. Whereas we find a slower fluctuation which may provide mild support for the imperfect substitution hypothesis.

4.3. *Confounding Events*

In order to determine whether there are any confounding events that could possibly be driving these results, we have extracted all news announcements disseminated on the RNS, for each company between 1993 and 1998. There are a total of 68 different categories of announcements, some of which are; mergers, acquisitions, joint ventures, rights issues, AGM, divestments, board changes, capital changes, corporate reorganisations, dividends and results. Elimination of all events with another announcement falling into the (-10,+10) window brings down the number of events analysed from 16,726 to 11,069.

[Please insert Figure 3 about here]

However, despite this elimination, the picture remains pretty much the same, as can be seen in Figure 3. The CARs for sells even exceed the 3% level after day 0. Therefore, there is not enough evidence to suggest that confounding events drive the results.

4.4. *Changes in Publication Rules*

To find out whether the behaviour of CARs could be affected by the different transparency rules in the Stock Exchange, we have split the data into 3 periods according to the date of the change in the trade delay rules. In Period 1 (pre-13 December 1993), the maximum allowable delay is 90 minutes, which is only available to trades larger than 3 times NMS. In Period 2 (13 December 1993 - 31 December 1995), the maximum allowable delay becomes 5 business days that is available for trades larger than 75 times NMS. Finally, in Period 3 (1 January 1996-onwards), although the maximum allowable delay remains the same, the lower bound is decreased from 90 minutes to 60 minutes for trades larger than 75 times NMS. However, the publication delay of trades between 3 and 6 times NMS was abolished.

[Please insert Figure 4 about here]

In Period 1, both buys and sells behave in a similar way before the announcement takes place. However, after the announcement of the trade, buys continue to yield positive abnormal returns, whereas there seems to be a downward price pressure for sells.

[Please insert Figure 5 and Figure 6 about here]

The general trend displayed by CARs in Periods 2 and 3 are similar to each other. However, in Period 1, the CARs for buys never manage to become positive despite positive abnormal returns between days -3 and 8. Overall, looking at the CARs, the change in trade publication rules does not seem to have much influence on the market response to buy and sell trades in Periods 2 and 3. However, CARs in Period 1 seem to behave differently than the other two periods.

4.5 *Do Institutional Investors behave homogeneously?*

In the results above, we have assumed that institutional investors are homogenous. However, previous studies show that some category of investors, such as banks, investment trusts and insurance companies, may be pressure sensitive while others, such as occupational pension funds, are pressure resistant (e.g., Brickley, Lease and Smith, 1988, 1994). We therefore, split our sample into 8 categories of institutional investors and compute for each the abnormal returns around the buy and sell trades. Table 5 reports the abnormal performance over the [-5 to +1] period by

category of institutional investor and size of the trade. Panel A reports the results for the sample as a whole and shows that all trades result in positive abnormal performance. However, the abnormal returns are not linearly related to the size of the trade. For pressure sensitive institutional investors (Panel B to Panel F), the abnormal performance is not most cases positive and significant. In contrast, for pension funds, the pressure resistant institutional investors, the abnormal performance is in most cases not significant.

[Please insert Table 5 about here]

The most striking results are reported in Table 6 where we show the behaviour of share prices around the announcement of the trades. Panel A reports the abnormal performance around the trades undertaken by fund managers. These investors appear to time perfectly their trades. Before they buy, i.e., in the 35-day pre-trade period, share prices decrease abnormally by 0.65%. Before they sell, shares prices increase abnormally by an average of 3.1%. In the post-event period [+2 to +40], share prices decrease by 0.24% for the buys and 0.02% for the sells, but this abnormal performance is not statistically significant. Similar timing of trades is observed for the case of investment trusts. The investment trusts do not appear to buy winners and sell losers as share prices carry on increasing by an average of 2.1% after the buy order while the post-abnormal performance of their sells is not statistically significant. In contrast, pension funds appear to lose on their trades. For example, in the pre-buy period [−40 to −6], share prices increase by 2.1% and then they decrease by 1.36%. On the event date 0, share prices increase for both buy and sell orders independently of the category of institutional investor.

[Please insert Table 6 about here]

5. Conclusions

In this paper we use a unique data set to analyse the market reaction to large block trades undertaken by a number of institutional investors in the UK over the period 1993 to 1998. A total of 8,590 buy orders and 8,136 sell orders are analysed. The results show that, on average, the market reacts positively to both sell and buy orders. However, we find significant differences in the trading patterns across institutional investors. We show that fund managers and investment trusts time their trades while pension funds adopt a momentum strategies as they buy after share price

run-up and sell when share prices decrease. Our results do not provide support for the monitoring hypothesis but suggest that block trades reflect the trading strategies of institutional investors and that some institutions are better than the others in timing their trading strategies.

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Figure 1
Abnormal Returns in Event Window (-40,+40)
 (OLS Market Model with Scholes & Williams (1977) Adjustment; N= 16726 events)

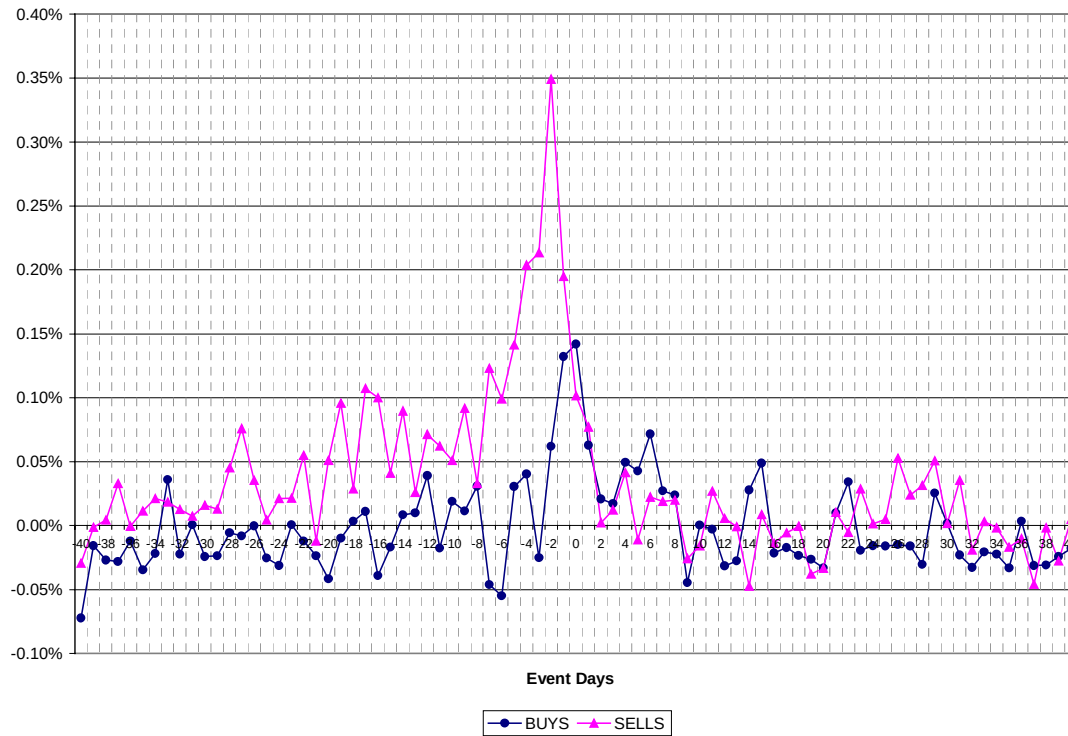


Figure 2
Cumulative Abnormal Returns for Event Window (-40,+40)
 (OLS Market Model with Scholes & Williams (1977) Adjustment; N= 16726 events)

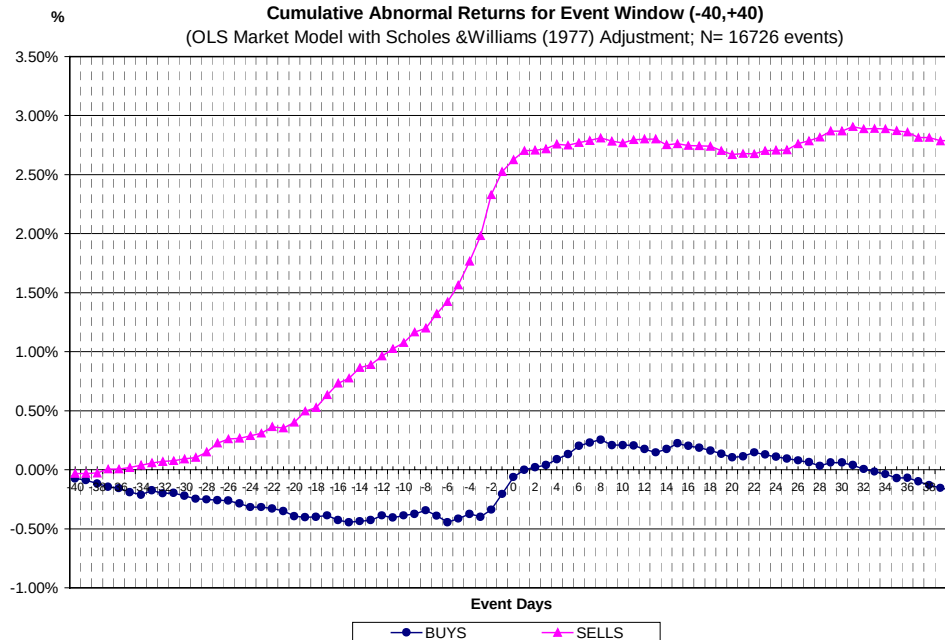


Figure 3
 % Cumulative Abnormal Returns excluding all confounding events in (-10,+10) window (N=11,069 events)

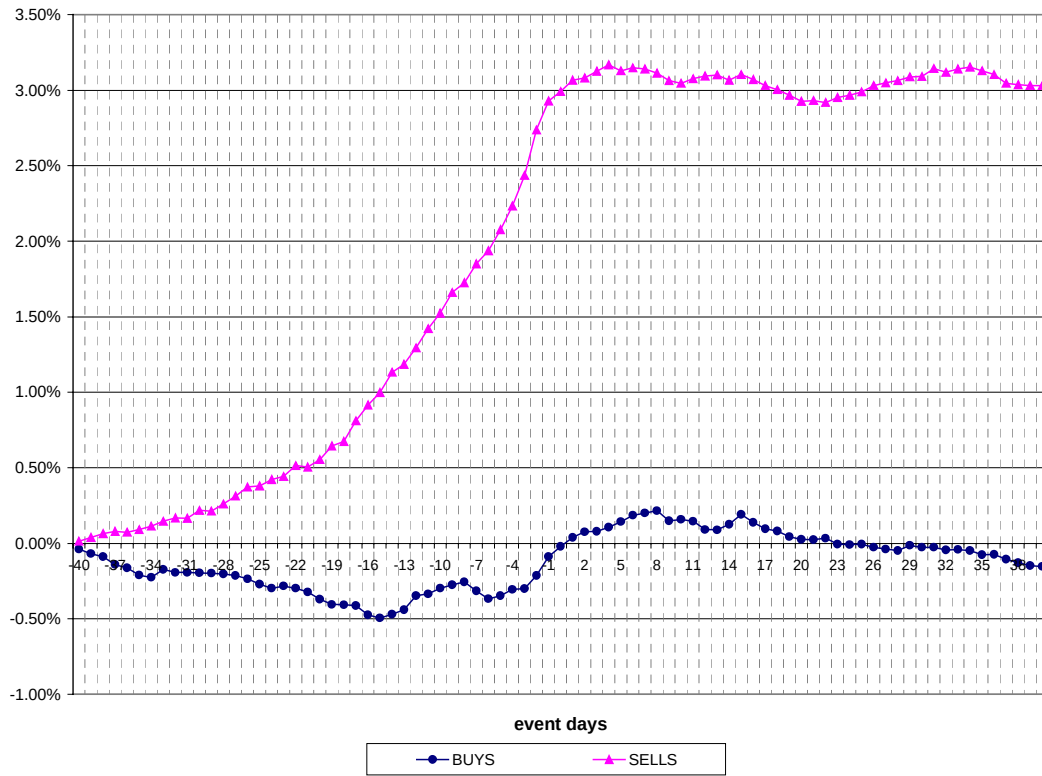
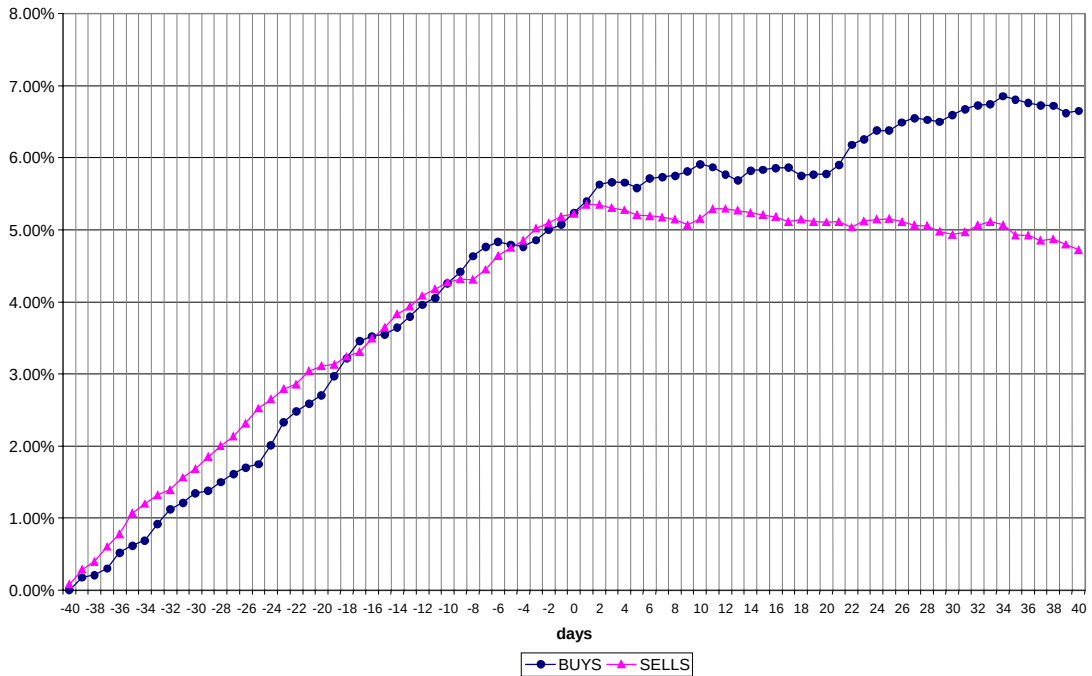


Figure 4
 % Cumulative Abnormal Returns- Period 1 (pre- 13 December 1993)



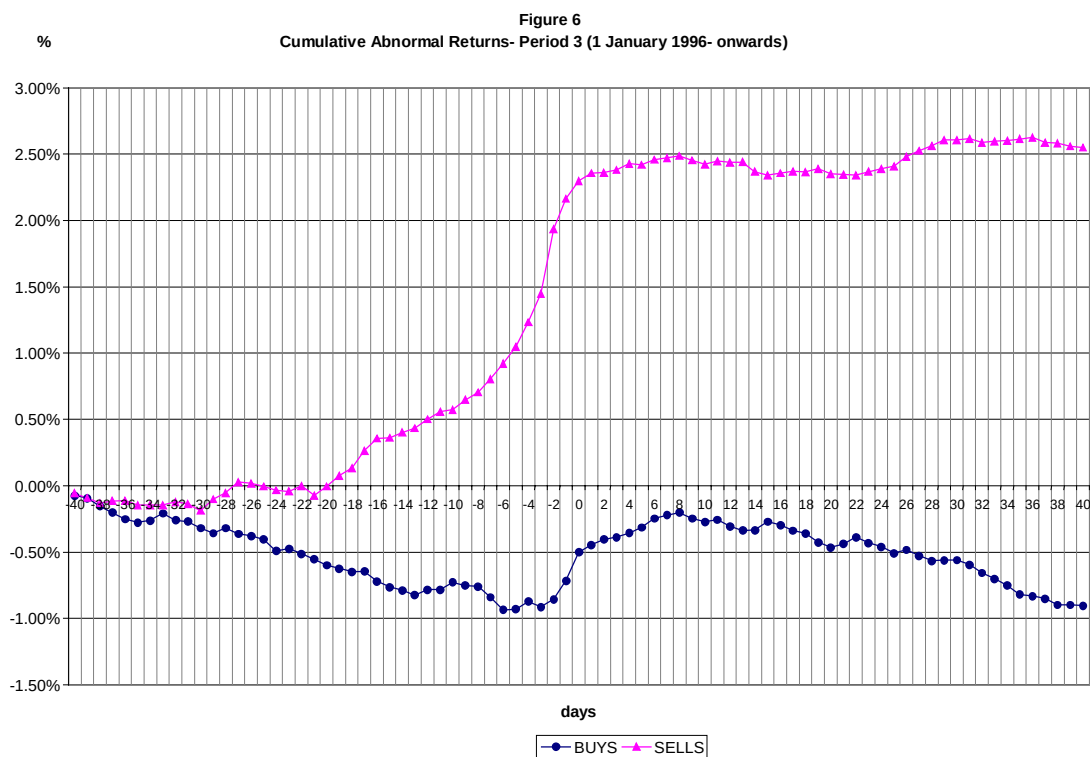
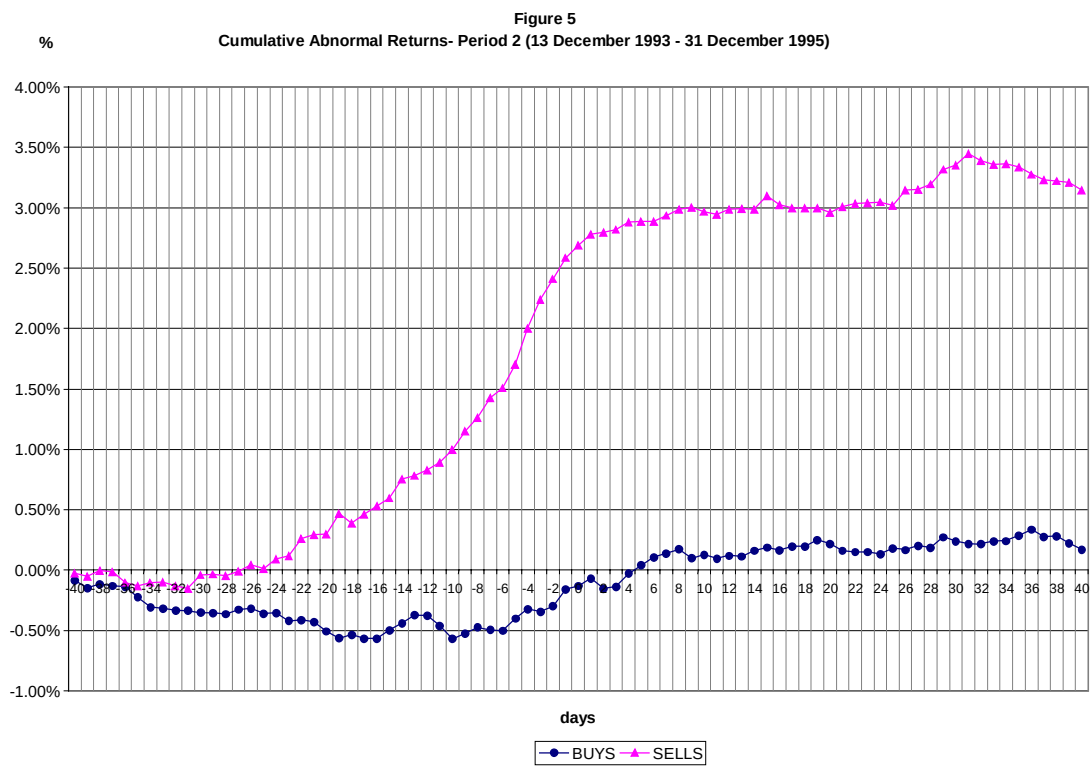


Table 1
Number and Size of Trades for the Sample Period (1993-1998)

	1993	1994	1995	1996	1997	1998	Total
Panel A: Number of Trades							
Buys	694	1001	1505	1767	1797	1826	8590
Sells	1063	814	1365	1596	1666	1632	8136
Total	1757	1815	2870	3363	3463	3458	16726
Panel B: Size of Trades							
Average % of Share Capital Traded	2.38%	3.10%	2.32%	2.18%	2.10%	1.99%	
Median % of Share Capital Traded	1.42%	1.26%	1.01%	0.97%	0.97%	0.93%	
Average £ Value of Trade (£'million)	11.1	11.6	17.2	13.0	11.1	9.5	
Median £ Value of Trade (£'million)	1.3	1.1	1.1	1.0	1.1	1.1	

Table 2
Descriptive Statistics for Trades

	Mean	Min	25th Percentile	Median	75th Percentile	Max
Panel A: All Trades						
% of share capital traded	2.27%	0.01%	0.37%	1.01%	3.02%	57.26%
% of shares held after the trade	8.32%	0.00%	3.92%	6.80%	12.10%	76.73%
Market Capitalisation (£'million)	616	0.03	37	133	493	51,897
£ value of the trade (£'million)	12.3	0.0	0.3	1.1	4.4	4,613.9
Panel B: Buys						
% of share capital traded	1.97%	0.01%	0.39%	0.96%	1.97%	57.26%
% of shares held after the trade	10.11%	2.86%	5.07%	9.08%	13.26%	76.73%
Market Capitalisation (£'million)	605	0.03	38	135	500	46,454
£ value of the trade (£'million)	10.8	0.0	0.3	1.0	4.1	3,354
Panel C: Sells						
% of share capital traded	2.58%	0.01%	0.34%	1.11%	3.40%	57.26%
% of shares held after the trade	6.43%	0.00%	0.00%	4.78%	10.84%	72.73%
Market Capitalisation (£'million)	628	0.03	36	132	487	51,897
£ value of the trade (£'million)	13.8	0.0	0.3	1.2	4.9	4,614

Table 3
Number of Trades for Each Type of Institutional Buyer / Seller

% of Share Capital Traded	<3%	3% to 5%	5% to 10%	10% to 15%	15% to 20%	>=20%	Total
Panel A: All Trades							
Fund Managers	5136	602	350	462	99	32	6681
Insurance Companies	2056	647	112	16	3	0	2834
Overseas Institutions	1595	237	96	27	6	21	1982
Banks	734	209	63	89	10	3	1108
Industrial & Commercial Companies	627	101	52	17	7	9	813
Investment Trusts	558	173	57	13	1	2	804
Pension Funds	373	192	54	7	3	3	632
All Other Institutions*	1285	356	116	97	10	8	1872
Total	12364	2517	900	728	139	78	16726
Panel B: Buys							
Fund Managers	2880	234	177	175	50	12	3528
Insurance Companies	1073	172	52	4	1	0	1302
Overseas Institutions	1024	78	41	14	3	11	1171
Banks	343	56	27	37	5	3	471
Industrial & Commercial Companies	437	37	16	4	3	4	501
Investment Trusts	384	49	18	4	1	1	457
Pension Funds	180	46	18	2	1	2	249
All Other Institutions*	704	102	54	43	4	4	911
Total	7025	774	403	283	68	37	8590
Panel C: Sells							
Fund Managers	2256	368	173	287	49	20	3153
Insurance Companies	983	475	60	12	2	0	1532
Overseas Institutions	571	159	55	13	3	10	811
Banks	391	153	36	52	5	0	637
Industrial & Commercial Companies	190	64	36	13	4	5	312
Investment Trusts	174	124	39	9	0	1	347
Pension Funds	193	146	36	5	2	1	383
All Other Institutions*	581	254	62	54	6	4	961
Total	5339	1743	497	445	71	41	8136

(*) All Other Institutions category includes trades by Life Assurance Companies, Charities Trusts and Foundations, Joint and Nominee Accounts, Other Financial Institutions and Groups, Venture Capital Companies, and the Public Sector.

Table 4
Abnormal Returns in Event Window (-40,+40)

(Using OLS Market Model with Scholes & Williams (1977) Adjustment; N=16726 events
Day 0 is the Announcement Day of the Trade; *:significant at 0.10 level, **:significant at 0.05 level,
***: significant at 0.01 level.)

Day	Buys	t-statistic	Sells	t-statistic	Day	Buys	t-statistic	Sells	t-statistic
-40	-0.07%	-1.094	-0.03%	-0.524	1	0.06%	0.947	0.08%	1.377
-39	-0.02%	-0.236	0.00%	-0.028	2	0.02%	0.317	0.00%	0.040
-38	-0.03%	-0.408	0.00%	0.078	3	0.02%	0.260	0.01%	0.220
-37	-0.03%	-0.427	0.03%	0.591	4	0.05%	0.749	0.04%	0.740
-36	-0.01%	-0.181	0.00%	-0.011	5	0.04%	0.644	-0.01%	-0.199
-35	-0.03%	-0.521	0.01%	0.204	6	0.07%	1.082	0.02%	0.400
-34	-0.02%	-0.330	0.02%	0.376	7	0.03%	0.410	0.02%	0.339
-33	0.04%	0.546	0.02%	0.330	8	0.02%	0.362	0.02%	0.356
-32	-0.02%	-0.340	0.01%	0.225	9	-0.04%	-0.672	-0.03%	-0.462
-31	0.00%	0.010	0.01%	0.134	10	0.00%	0.008	-0.02%	-0.282
-30	-0.02%	-0.368	0.02%	0.287	11	0.00%	-0.045	0.03%	0.480
-29	-0.02%	-0.359	0.01%	0.228	12	-0.03%	-0.475	0.01%	0.100
-28	-0.01%	-0.084	0.05%	0.808	13	-0.03%	-0.416	0.00%	-0.012
-27	-0.01%	-0.121	0.08%	1.355	14	0.03%	0.421	-0.05%	-0.846
-26	0.00%	-0.003	0.04%	0.632	15	0.05%	0.737	0.01%	0.155
-25	-0.03%	-0.383	0.00%	0.081	16	-0.02%	-0.324	-0.01%	-0.247
-24	-0.03%	-0.472	0.02%	0.377	17	-0.02%	-0.258	-0.01%	-0.099
-23	0.00%	0.010	0.02%	0.381	18	-0.02%	-0.354	0.00%	-0.010
-22	-0.01%	-0.182	0.05%	0.980	19	-0.03%	-0.398	-0.04%	-0.679
-21	-0.02%	-0.356	-0.01%	-0.215	20	-0.03%	-0.499	-0.03%	-0.598
-20	-0.04%	-0.626	0.05%	0.909	21	0.01%	0.147	0.01%	0.185
-19	-0.01%	-0.151	0.10% *	1.713	22	0.03%	0.516	-0.01%	-0.096
-18	0.00%	0.050	0.03%	0.516	23	-0.02%	-0.290	0.03%	0.512
-17	0.01%	0.167	0.11% *	1.920	24	-0.02%	-0.237	0.00%	0.023
-16	-0.04%	-0.592	0.10% *	1.784	25	-0.02%	-0.241	0.00%	0.087
-15	-0.02%	-0.258	0.04%	0.731	26	-0.01%	-0.223	0.05%	0.943
-14	0.01%	0.128	0.09%	1.604	27	-0.02%	-0.240	0.02%	0.427
-13	0.01%	0.148	0.03%	0.464	28	-0.03%	-0.456	0.03%	0.560
-12	0.04%	0.590	0.07%	1.276	29	0.03%	0.385	0.05%	0.907
-11	-0.02%	-0.264	0.06%	1.113	30	0.00%	0.024	0.00%	0.032
-10	0.02%	0.286	0.05%	0.911	31	-0.02%	-0.349	0.04%	0.633
-9	0.01%	0.173	0.09%	1.638	32	-0.03%	-0.496	-0.02%	-0.340
-8	0.03%	0.466	0.03%	0.591	33	-0.02%	-0.310	0.00%	0.058
-7	-0.05%	-0.697	0.12% **	2.202	34	-0.02%	-0.340	0.00%	-0.033
-6	-0.05%	-0.829	0.10% **	1.771	35	-0.03%	-0.500	-0.02%	-0.305
-5	0.03%	0.460	0.14% **	2.525	36	0.00%	0.052	-0.01%	-0.182
-4	0.04%	0.611	0.20% ***	3.644	37	-0.03%	-0.473	-0.05%	-0.818
-3	-0.03%	-0.382	0.21% ***	3.815	38	-0.03%	-0.469	0.00%	-0.033
-2	0.06%	0.937	0.35% ***	6.241	39	-0.02%	-0.367	-0.03%	-0.494
-1	0.13% **	1.998	0.19% ***	3.482	40	-0.02%	-0.272	0.00%	0.088
0	0.14% **	2.143	0.10% **	1.816					

Table 5
Cumulative Abnormal Returns (-5,+1) for Major Institutional Categories
(OLS Market model with Scholes & William (1977) Adjustment)
(*:significant at 0.10 level, **:significant at 0.05 level,***: significant at 0.01 level)

	SIZE OF THE CHANGE IN SHARE STAKE					
	0 to 3%	3 to 5%	5 to 10%	10 to 15%	15 to 20%	20% and more
Panel A: All Institutions						
All Trades	0.659% ***	1.464% ***	1.528% ***	1.202% ***	0.487% ***	1.442% ***
Buys	0.340% ***	1.085% ***	0.838% ***	0.988% ***	-0.887% ***	1.208% ***
Sells	1.079% ***	1.645% ***	2.092% ***	1.277% ***	1.776% ***	1.771% ***
Panel B: Fund Managers						
All Trades	0.773% ***	1.341% ***	0.794% **	0.889% ***	-0.297%	-0.165%
Buys	0.453% ***	1.360% ***	0.719% *	0.645%	-1.005%	-0.833%
Sells	1.181% ***	1.329% ***	0.871% *	1.038% ***	0.425%	0.235%
Panel C: Insurance Companies						
All Trades	0.523% ***	1.045% ***	2.697% ***	0.639%	8.042% **	n/a
Buys	0.318% **	0.581% *	2.081% ***	1.030%	-0.428%	n/a
Sells	0.747% ***	1.214% ***	3.230% ***	0.508%	12.277%	n/a
Panel C: Overseas Institutions						
All Trades	0.426% ***	0.488%	2.827% ***	2.143%	6.835%	0.072%
Buys	-0.085%	-0.035%	0.946% ***	3.714% ***	-2.884% ***	0.170%
Sells	1.342% ***	0.743% *	4.264% ***	0.450%	16.554% *	-0.036%
Panel D: Banks						
All Trades	0.502% ***	1.274% ***	0.237%	-0.022%	-1.166%	-2.049%
Buys	0.391%	2.031% ***	1.430%	-0.692%	-1.237%	-2.049%
Sells	0.599% **	0.998% **	-0.658%	0.455%	-1.095%	n/a
Panel E: Industrial and Commercial Companies						
All Trades	0.592% ***	1.597% ***	1.052%	-1.353%	5.577%	8.640% **
Buys	0.245%	2.782% ***	0.170%	5.435%	4.641%	11.613% *
Sells	1.391% ***	0.912%	1.444%	-3.442% *	6.280%	6.261%
Panel F: Investment Trusts						
All Trades	1.616% ***	4.827% ***	4.038% ***	3.058% **	1.085%	7.895%
Buys	1.323% ***	3.227% ***	0.671%	1.971%	1.085%	2.284%
Sells	2.264% ***	5.460% ***	5.591% ***	3.541% *	n/a	13.505%
Panel G: Pension Funds						
All Trades	0.133%	1.627% ***	0.451%	2.099%	0.519%	7.524%
Buys	-0.021%	-1.188%	-0.334%	-2.640%	0.972%	3.836%
Sells	0.277%	2.515% ***	0.844%	3.995% **	0.293%	14.902%

Table 6
Selected Abnormal Return Measures for Major Institutional Categories
(OLS Market model with Scholes & William (1977) Adjustment)
(*:significant at 0.10 level, **:significant at 0.05 level,***: significant at 0.01 level)

	DAY 0	CAR(-40,-6)	CAR(-5,+1)	CAR(+2,+40)
Panel A: Fund Managers				
Buys	0.133% ***	-0.647% ***	0.511% ***	-0.246%
Sells	0.079% **	3.134% ***	1.152% ***	-0.016%
Panel B: Insurance Companies				
Buys	0.118% ***	-0.527% **	0.425% ***	-0.858% ***
Sells	0.083% *	1.376% ***	0.998% ***	1.580% ***
Panel C: Overseas Institutions				
Buys	0.184% ***	-0.688% *	-0.004%	0.027%
Sells	0.138% *	-1.091% **	1.464% ***	-0.113%
Panel D: Banks				
Buys	0.088%	1.136% *	0.528% **	-1.460% **
Sells	0.018%	-0.667%	0.600% ***	-1.492% ***
Panel E: Industrial and Commercial Companies				
Buys	0.405% ***	-1.005% **	0.588% ***	-1.298% ***
Sells	0.075%	0.248%	1.215% ***	1.408% *
Panel F: Investment Trusts				
Buys	0.278% ***	-0.099%	1.509% ***	2.066% ***
Sells	0.306% **	2.762% ***	3.801% ***	-0.277%
Panel G: Pension Funds				
Buys	0.046%	2.069% ***	-0.246%	-1.358%
Sells	0.269% ***	-0.644%	1.260% ***	-1.450% **

¹ Scholes (1972) quotes (in footnote 31) the Report of Special Study of Securities Markets of the Securities and Exchange Commission (1963): “ An official of a large mutual fund selling organization stated to the study that the funds sponsored by it sometimes used secondary distributions to dispose of ‘sick’ situations rapidly” to support his point (p.201).

² The Normal Market Size (NMS), as defined by the London Stock Exchange, is 2.5 % of the average daily number of shares traded in the stock over the past year. Stocks with an NMS of at least 2,000 shares are classified as more liquid and stocks with an NMS less than 2,000 shares are classified as less liquid. The NMS Multiple equals the trade size divided by the NMS.

³ The 5% threshold follows the Securities and Exchange Commission’s filing procedures.

⁴ They define ‘normal’ trading volume as the median daily volume for the previous 21 days. They use a fairly short period in this definition due to the seasonality characteristics of volume.

⁵ Firm size is measured in terms of the quintile of the average month-end market value of the firm’s equity during 1982.

⁶ Since money managers’ performance evaluation is usually based on ‘peer group’ comparisons, it could be tempting to invest in a similar portfolio as other money managers. By this way, a money manager could avoid extreme under-performance at a cost to the client.

⁷ In simple terms, contrarian strategies are based on buying currently unpopular assets. Therefore, the short-term performance of these stocks might be appalling, especially when compared to a popular benchmark such as the FTSE-100 Index. Opponents of this strategy argue that it may put excessive strain on the fund manager and consequently lead to short-termism.

⁸ The permanent effect depends on the size of the block (Mikkelson and Partch (1985)).

⁹ The corresponding figures in their 1993 paper are 0.34% for buy and –0.04% for sell-packages.

¹⁰ Following Maug (1998), the word “monitoring” is used here to encompass all value enhancing activities, including intervention in company’s operations and information acquisition to pinpoint possible problem areas that may need intervention. Accordingly, “monitoring” is used synonymously with “intervention” and “shareholder activism”.

¹¹ Their definition of an activist blockholder is those investors “who announce their intention of influencing firm policies or who are known for activist policies in the past” (p.611).

¹² They calculate ROA as earnings before interest, taxes, depreciation and amortisation (EBITDA) divided by assets.

¹³ Views expressed by Nicola Humpage, London Stock Exchange at the Expo’99, 23 November 1999, London.

¹⁴ For the calculation of this measure, please refer to footnote 2.

¹⁵ In 1996, trades greater than 6 times NMS make up 2% by number and 34% by value of all trades in the market.

¹⁶ They define a buy trade package as successive purchases of the same stock by the same money manager with a break of less than 5 days in between the individual trades and the event dates they use are based on actual trade dates. However, in my sample, day 0 corresponds to the announcement day, which can actually be 2 days after the trade took place, due to the legal notification period allowed for the shareholders to inform the company of the trade. Hence, if we could determine the exact trade date, these results might resemble each other even more.